

AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
771 A	44.08 MVA	90.0 degC	70.2 degC	61.524 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		240	mm2
Conductor type		stranded	
Conductor diameter Dc		18.4	mm
Conductor screen OD Dc_sc		19.4	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		34.8	mm
Ins outer screen OD Di_sc		35.8	mm
Insulation thickness t_ins		7.70	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
Screen type		Cu_wire	
Screen mean diameter ds		37.7	mm
Screen area As		35.6	mm2
Armour type		none	
Oversheath material		PE	
Cable outer diameter De		44.0	mm
Rated voltage U		33.0	kV
U0 line-to-earth		19.052	kV
Frequency		50	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		1000	mm
Axial spacing s		44.0	mm
Ambient ground temp theta_a		20	degC
Soil thermal resistivity rho_soil		1.00	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		7.5400e-05	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 20 degC		7.5400e-05	ohm/m
R0 at theta_max = 90 degC		9.6143e-05	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant		1.0000	
kp proximity constant		0.8000	
$xs4 = [ks(8\pi i f / R0) \times 10^{-7}]^2$		1.708396	
ys skin effect factor		0.008835	
$xp4 = [kp(8\pi i f / R0) \times 10^{-7}]^2$		1.093373	
yp" intermediate proximity		0.005669	
yp proximity factor		0.004298	
R at theta_max AC = $R0(1+ys+yp)$		9.7405e-05	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.004000	
U0 line-to-earth		19.0520	kV
Capacitance C = $er / (18 \ln(Di_sc / Dc_sc)) \times 10^{-9}$		0.226693	nF/m
$\omega = 2 \pi f$		314.1593	rad/s
$Wd = \omega C U0^2 \tan(\delta)$		1.0340e-01	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		35.63	mm2
Screen Rs0 at 20 degC		4.8389e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins		3.50	K.m/W
rho_osh		3.50	K.m/W
$T1 = (\rho_{ins} / 2\pi) \ln(Di_sc / Dc_sc)$		0.341286	K.m/W
T2 filler/bedding single-core = 0		0.000000	K.m/W
$T3_raw = (\rho_{osh} / 2\pi) \ln(1 + 2t/ds)$		0.061491	K.m/W
T3 x 1.6 factor touching trefoil		0.098385	K.m/W
$u = 2L/De$		45.4545	
$T4 = (\rho / 2\pi) \ln(u + \sqrt{u^2 - 1})$		0.717747	K.m/W
Sum T = T1+T2+T3+T4		1.157418	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		70.2120	degC
Rs at theta_s		5.7938e-04	ohm/m
X sheath reactance		5.3261e-05	ohm/m
lam1' circulating current		0.049845	
lam1" eddy current		0.010434	
lam1 = lam1' + lam1"		0.060279	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		70.00	K
Etop numerator		69.897966	
Ebot denominator		1.1753e-04	
I = sqrt(Etop/Ebot) exact		771.1827	A
I rounded TB880 GP2		771	A
MVA = sqrt(3) x U x I		44.079	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.0000	degC
theta_s sheath surface		70.2120	degC
theta_j cable outer surface		64.1589	degC
theta_a ambient ground		20.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I^2 x R		57.929044	W/m
Sheath Ws = lam1 x Wc		3.491895	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		1.0340e-01	W/m
Total		61.524341	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.169535	uH/m
Reactance X = omega L		53.2611	uOhm/m
Pos-seq resistance R1		103.2766	uOhm/m
Pos-seq impedance Z1		116.2016	uOhm/m
Capacitance C		0.226693	nF/m
Charging current Ic		1.3568	mA/m